

Administrator Manual

Attendance Management System — Version 1.0.0

This manual covers installation, licensing, multi-company setup, user and permission management, backups, and troubleshooting — everything beyond day-to-day use, which is covered in the User Manual.

1. Installation

Run Setup.exe on the target Windows machine and follow the wizard:

- 1 Choose an install location (defaults to Program Files) — administrator rights are required for this one-time step.
- 2 Optionally check "Create a desktop icon."
- 3 Click Install, then Finish (the app can launch immediately from the final wizard page).

The installer creates Start Menu and (optionally) Desktop shortcuts, and registers an uninstaller under Windows' "Apps & Features." No Python installation or other runtime is required on the target machine — the installer bundles everything the application needs.

Portable use: Portable.exe (also available in the release) is a single, standalone executable — copy it to a USB drive or any folder and run it directly, with no installation step. It stores its data in the same %LOCALAPPDATA%\AttendanceManagementSystem\ location as the installed version when run on the same Windows user account.

1.1 Where application data lives

The application never writes inside its own install folder (Program Files is not writable by a standard user, and the goal is also to survive upgrades cleanly). All writable data — the database, logs, backups, uploaded files, and the license — lives under:

```
%LOCALAPPDATA%\AttendanceManagementSystem\  
├─ data\      (SQLite database)  
├─ logs\      (application logs, useful for troubleshooting)  
├─ backups\   (database backups)  
└─ uploads\   (employee photos, generated QR codes/barcodes, report exports)
```

This folder is created automatically on first launch and is never touched by installing, upgrading, or uninstalling the application — see "Upgrading" and "Uninstalling" below.

1.2 Upgrading

Running a newer Setup.exe over an existing installation performs an in-place upgrade: program files under the install directory are replaced, and %LOCALAPPDATA%\AttendanceManagementSystem\ (database, license, logs, backups) is left completely untouched. No manual data migration step is needed for a standard upgrade.

1.3 Uninstalling

Uninstall from Windows Settings → Apps, or via the Start Menu shortcut. The uninstaller removes only the installed program files. It then asks separately whether to also delete your data folder (%LOCALAPPDATA%\AttendanceManagementSystem\) — choose No to keep your database, license, and logs in place for a future reinstall, or Yes to fully remove all traces of the application and its data from the machine.

2. Licensing

The application requires an activated license before the login screen is shown. License keys are cryptographically signed (Ed25519) and bound to the specific machine they're activated on — a key generated for one computer will not activate on another.

2.1 Activation

On first launch (or whenever no valid license is present), the activation screen appears. Enter the license key you were provided and click Activate. On success, the application proceeds to the login screen; every subsequent launch skips this screen as long as the license remains valid.

2.2 License types

Type	Behavior
Trial	Time-limited; expires after its trial period.
Monthly / Yearly	Subscription-style; renewable, expires on the configured date.
Lifetime	Never expires.

2.3 Checking license status

The License Info screen (accessible from within the application once logged in — check the Settings area or your user menu) shows the current license type, expiry date (if any), and days remaining. Renew before expiry to avoid an interruption — contact your software vendor for a new key when a subscription is nearing its end.

2.4 Reactivation / transferring to a new machine

License keys are bound to the machine they were activated on. If you need to move the software to new hardware, contact your vendor for a reactivation — this application does not generate license keys itself; key generation is a vendor-side process kept separate from the shipped product for security (the signing key is never distributed with the application).

3. Multi-Company Setup

This is a true multi-tenant application: one installation, one database, any number of independent companies, each fully isolated from the others (no employee, department, user, device, or report from one company is ever visible to another). To onboard a new company, use the company management flow reachable from the login screen / initial setup — no code

changes or per-company deployment is required.

Each company has its own:

- Employees, departments, and shifts.
- Users, roles, and permissions.
- Devices and attendance records.
- Settings (time zone, date format, currency, backups).

4. Users, Roles & Permissions

- Roles are per-company and fully customizable — there is no fixed global role list beyond what you define.
- Permissions are granted per role from a fixed catalog covering every module (dashboard, employees, departments, attendance, devices, reports, users/roles, settings, backups), each typically split into .view and .manage (or .export) grades — e.g. a role can be allowed to view attendance without being able to edit it, or view reports without being able to export them.
- Audit log: every sensitive operation (logins, record changes, backups, permission changes) is recorded with who performed it and when, for accountability and troubleshooting.

To create a new user: go to Users → Add, assign a company (implicit — you manage users within the company you're logged into), a username, an initial password, and a role. The user's password is hashed (bcrypt) before storage; there is no way to view a user's existing password, only to reset it.

4.1 Account lockout & sessions

Accounts lock temporarily after repeated failed login attempts, to resist password-guessing. Sessions expire automatically after a period of inactivity, requiring the user to log in again — this cannot be disabled from the UI and is a deliberate security default.

5. Devices

Supported protocols: ZKTeco (TCP/UDP push/pull protocol) and Hikvision (ISAPI over HTTP). For each device you configure its IP address and port; use "Test Connection" before saving to confirm reachability. Punch synchronization pulls new records from the device; pushing employee data enrolls them for recognition on the device's own hardware (biometric/card enrollment itself always happens on the device, not in this application).

Network requirements: the computer running this application must be able to reach each device's IP/port — same subnet, or routed and allowed through any firewalls in between. If a device intermittently drops, check for DHCP lease changes on the device (a static IP or DHCP reservation on the device is strongly recommended).

6. Backup & Restore

- Manual backup: from Settings → Backup, create a backup on demand. Backups are SQLite-WAL-compatible database snapshots stored under %LOCALAPPDATA%\AttendanceManagementSystem\backups\.
- Restore: from the same screen, restore from a previously created backup file. Restoring replaces the current database — make a fresh backup of the current state first if you might need to go back to it.
- Recommended practice: back up before any major change (bulk employee import, department restructuring, a software upgrade) and keep periodic backups copied off-machine (network share, cloud storage) in case of hardware failure — the application's own backup feature protects against data mistakes, not against the machine itself being lost.

7. Troubleshooting

Symptom	Likely cause / fix
App won't start, no window appears	Check %LOCALAPPDATA%\AttendanceManagementSystem\logs\ for the most recent log file and the error it recorded.
"تعذر الاتصال بقاعدة البيانات" (database connection error) at startup	The data folder or database file may be missing/corrupted, or in use by another process. Restore from a recent backup if needed.
Device shows disconnected	Confirm the device is powered on and reachable (ping its IP); confirm the IP/port configured in Devices still matches the device.
Arabic text garbled in an exported PDF	Should not occur in a correctly built release — the app bundles its own Arabic-capable font and text-shaping libraries. If it does, the installation may be corrupted; reinstall.
Login screen shows no companies	No company has been set up yet, or all companies are marked inactive — check company setup / reactivate the company.
User locked out	Wait for the lockout window to pass, or have another administrator reset the account.
License shows expired	Contact your vendor for a renewed key and re-activate via the activation screen.

For issues not covered here, collect the relevant log file(s) from %LOCALAPPDATA%\AttendanceManagementSystem\logs\ before contacting support — they contain the detail needed to diagnose most problems quickly.